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AP Assignment 1

Description of my code:

This is a simple course registration system. Students can see available courses, sign up for courses, drop courses, and submit complaints. Professors can manage their courses and view the students enrolled. Administrators manage the list of courses, student records, complaints. This is all done by using OOPs basic concepts like Encapsulation, Interfaces, Inheritance, Polymorphism etc.

Assumptions in my code:

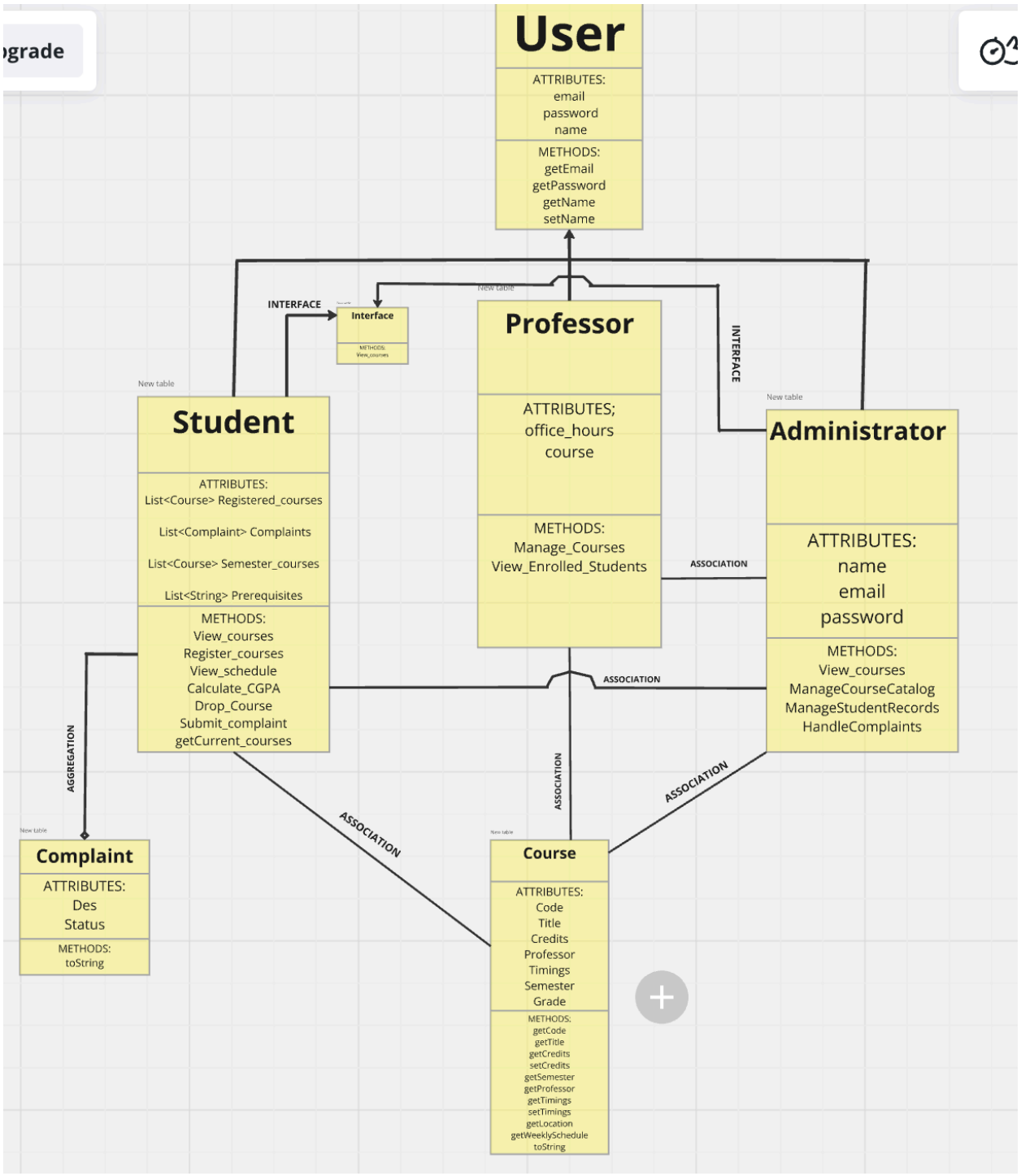
- I am taking only 3 semesters for example and I have defined a few courses in each semester.
- I have created 3 Student objects, 6 Professor objects, 1 Admin object
- Each professor can have only one course.
- Log in details asked in the main function for which log in email and password are fixed for each student and professor, and a fixed password for admin.
- Input Output format is the same as asked in the assignment.

OOPS concepts that have be applied:

Classes: Used several classes - Student, Professor, Administration, Complaint, User, Course.
Abstraction: Used an abstract class User where it declares methods inherited by other classes.
Encapsulation: Uses private fields with public getters and setters to protect and access data to other classes.
Inheritance: Class Student, Professor and Administrator have used inheritance where they inherit common properties from User class.
Interface: Uses Interface which has a method declared as View_courses() implemented by both Student and Administrator class as they both have to display the courses.
Polymorphism: Overrides the toString method from Object class in multiple files to fit the printing better.
Aggregation: Complaint linked to Student where students can submit complaints.
Association: The Professor, Administrator, Student associated with the course class.

MIRO LINK FOR UML:
https://miro.com/app/board/uXjVLcoMjs0=/

UML DIAGRAM:


User
ATTRIBUTES:
email
password
name
METHODS:
getEmail
getPassword
getName
setName
INTERFACE
Interface
Professor
ATTRIBUTES;
office_hours
course
METHODS:
Manage_Courses
View_Enrolled_Students
INTERFACE
Student
ATTRIBUTES:
List<Course> Registered_courses
List<Complaint> Complaints
List<Course> Semester_courses
List<String> Prerequisites
METHODS:
View_courses
Register_courses
View_schedule
Calculate_CGPA
Drop_Course
Submit_complaint
getCurrent_courses
Administrator
ATTRIBUTES:
name
email
password
METHODS:
View_courses
ManageCourseCatalog
ManageStudentRecords
HandleComplaints
ASSOCIATION
ASSOCIATION
AGGREGATION
ASSOCIATION
ASSOCIATION
Complaint
ATTRIBUTES:
Des
Status
METHODS:
toString
Course
ATTRIBUTES:
Code
Title
Credits
Professor
Timings
Semester
Grade
METHODS:
getCode
getTitle
getCredits
setCredits
getSemester
getProfessor
getTimings
setTimings
getLocation
getWeeklySchedule
toString